SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – 25093**
- **Problem Statement Title- Centralised Digital Platform for Comprehensive student activity record in HEIs**
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID- 60538**
- **Team Name - CodeWiz Crew**

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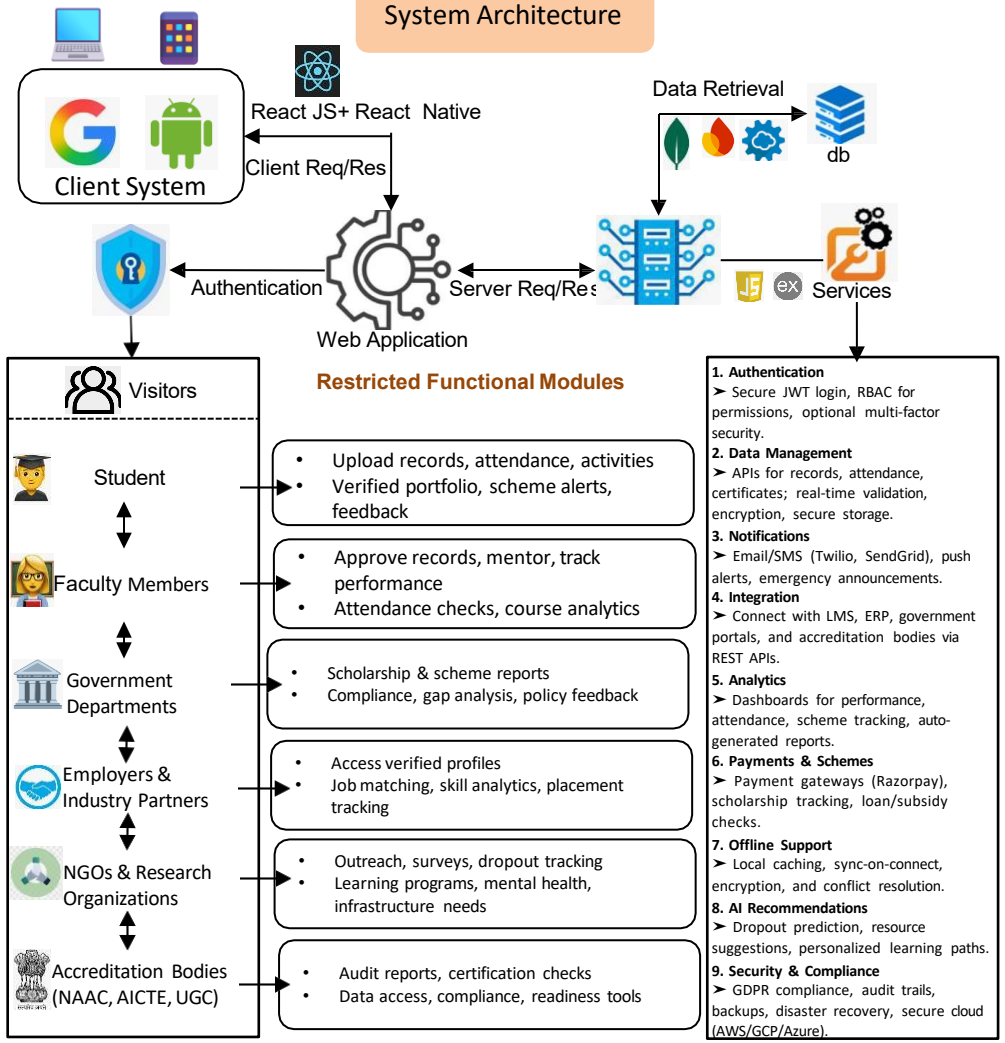
❖ Proposed Solution

1. **Verified Records & Portfolios** – **Structured upload** of **academic**, **cultural**, **technical**, and **internship** achievements with **metadata**; **institution-approved PDFs**; **QR-secured blockchain** certificates; and **faculty/admin validation** for authenticity.
2. **Attendance, Tracking & Analytics** – **Real-time attendance** and **performance monitoring** with **offline caching**, **SMS alerts** for low-connectivity regions, and **interactive dashboards** for institutional and **policy-level decision-making.**
3. **Accreditation & Compliance Automation** – **NAAC-ready SSR templates**, **surveys**, and **peer-review tools**, with integrated **data reporting** for **NAAC**, **AICTE**, and **government compliance standards.**
4. **Offline, Inclusive & Adaptive Access** – **Offline data entry**, **SMS updates**, and **community kiosks**; **gender-inclusive features** like **anonymous reporting**, **safe access**, and **local language UI**; plus **PWA/device-agnostic apps** for rural, shared, and low-bandwidth environments.
5. **AI-Powered Learning Insights** – **Personalized recommendations**, **mentoring support**, and **career planning** powered by **AI/ML analytics** for skill development and **employability**.
6. **Teacher, Leadership & Institutional Tools** – **Digital training**, **pedagogy enhancement**, **governance dashboards**, and alignment with **NEP's 5-3-3-4 structure** for skill-based, stress-reduced education.
7. **Schemes, Scholarships & Opportunities** – Integrated access to **scholarships**, **fellowships**, **financial aid**, and **employment programs** to enhance **socio-economic support** for students.
8. **Security, Continuity & Institutional Trust** – **Cloud-based**, **containerized setup** with **SSL**, **RBAC**, **GDPR** compliance, and **audit trails**; **conflict-resilient offline workflows** and **alerts**; and **tamper-proof verified achievements** that enhance institutional credibility, employability, and mobility.

System Architecture



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TECHNICAL APPROACH

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❖ Technology Stack

APIs :
- REST, GraphQL,
- SMS/Email (Twilio, SendGrid),
- Geo, Weather, OCR
- Blockchain Timestamp
- Payment Gateways (Razorpay),
- Authentication APIs

Security
- SSL, GDPR, RBAC,
- Encryption, Backups
- Audit Trails, Blockchain Verification, MFA

Cloud & Deployment
- AWS, GCP, Azure
- Docker, Kubernetes
- Terraform, CI/CD
- NGINX, Cloud
- Storage, CDN

Analytics
- Power BI, Tableau, Kafka
- Spark, Google Analytics
- BigQuery, Metabase,
- SSR & Feedback Reporting

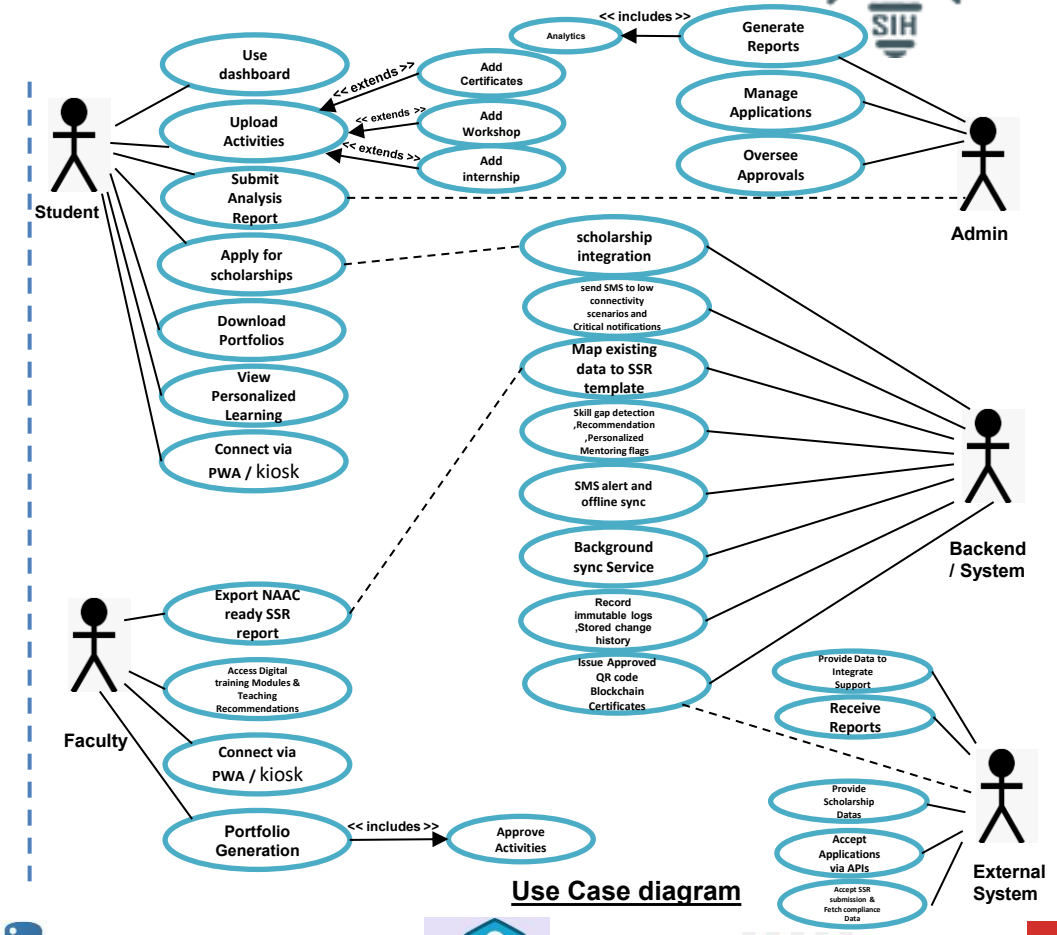
Student Hub
- Academic Guidance
- Performance Analytics
- Dropout predictions
- Learning recommendations

AI / ML services
- ChatGPT (GPT-5 mini)
- Python, scikit-learn
- TensorFlow, Transformers
- Time-Series ML, Learning
- Analytics, Personalized Recommendations

Backend
- Node.js, Express, Django,
- MongoDB, Firebase,
- REST, JWT, Oauth, RBAC
- API versioning
- Rate limiting, Redis

Front-End
- React.js, React Native, React Router
- Tailwind, Redux, PWA
- WebSocket, Offline Storage
- i18n, Voice Assistance

Offline & Edge
- Service Workers, IndexedDB
- SMS Alerts, Community Kiosks
- Offline Sync



Use Case diagram

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FEASIBILITY AND VIABILITY

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⚖️ Feasibility :

- Integrates academic, extracurricular, and skill data from day one into structured profiles, reducing paperwork and enhancing transparency.
- Supports offline submissions, SMS alerts, and community kiosks to tackle internet shutdowns and curfews.
- Aligns with NAAC accreditation criteria for seamless reporting and evaluation.
- Encourages participation across genders and regions by offering local language interfaces and safe access pathways.

⚙️ Viability :

1. Empowers students with verified portfolios, improving job and higher education access.
2. Helps institutions meet accreditation requirements, reducing administrative burden and enhancing governance
3. Supports distance learning, mentoring, and skill development amid teacher shortages and infrastructure gaps.
4. Builds trust through institution-sealed records, improving student engagement and career readiness.

⚠️ Challenges :

1. Frequent curfews and internet cuts disrupt learning.
2. Rural infrastructure gaps limit device access.
3. Safety concerns and gender barriers lower attendance.
4. Outdated teaching reduces learning outcomes.
5. Data collection delays complicate accreditation.

Solutions :

1. Centralized verified profiles with approval workflows.
2. Offline-first design with SMS, kiosks, and caching.
3. Inclusive interfaces in local languages with voice assistance.
4. NAAC-aligned analytics for audits and reports.
5. AI recommendations for learning and mentoring.

📊 Use Cases :

1. Unified platform for academic, internships, and activities.
2. Offline support for remote learners.
3. Verified portfolios for placements and admissions.
4. Scholarship and government reporting made easy.
5. Encourages female education through secure access.

📈 Business Potentials :

1. Subscription models for institutions and policy bodies.
2. Partnerships with edtech and training providers.
3. Data tools for counseling, scholarships, and governance.
4. Expands education in conflict and rural zones.
5. Scalable to other underserved regions.

⭐ Supporting Facts for Feasibility and Viability ⭐

- Literacy in J&K is **68.74%**, with **female literacy ~58%**, highlighting the need for safe, inclusive education tools.
- **100+ days/year** of internet shutdowns and **30%+ lost school days** show offline-first access is critical.
- **60% of rural schools** lack classrooms/sanitation; **45% lack internet**, limiting digital learning access.
- **35%+ teacher vacancies** and outdated methods reduce outcomes, needing data-driven mentoring.
- **70% of institutions** face documentation gaps in NAAC audits, requiring verified records and automated reports.
- **40%+ students** miss scholarships due to paperwork and lack of digital access, stressing the need for seamless integration.
- **100,000+ learners** rely on distance education in remote areas, needing structured tracking and analytics.
- NEP 2020's skill-focus demands infrastructure support in conflict-affected, rural regions.
- Safety concerns push girls out of school; local language UI and anonymous reporting improve access.

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IMPACT AND BENEFITS

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❖ Benefits of the solution :

Social

•Empowers students with **verified profiles**, boosting **trust** and **transparency**.
•Increases **inclusion** through **local language UI** and **offline access**.
•Supports **gender equity** via **safe, anonymous reporting tools**.

Economical

•Enhances **employability** with **accredited portfolios** for **jobs** and **higher studies**.
•Simplifies **scheme access**, reduces **paperwork**, and enables timely **scholarships**.
•Strengthens **rural economies** through improved **education outcomes** and **skill development**.

Educational

•Delivers **personalized learning analytics** for **performance improvement**.
•Facilitates **mentorship** with **faculty-driven, data-backed guidance**.
•Bridges **infrastructure gaps** using **mobile, offline, and assistive tools**.

Technological

•Leverages **AI** for **attendance tracking**, **portfolio validation**, and **predictive analytics**.
•Ensures **secure, scalable access** with **blockchain-style verification** and **data encryption**.
•Integrates seamlessly with **existing LMS, ERP, and government systems**.

Environmental

•Reduces **paper-based records**, promoting **digital** and **eco-friendly processes**.
•Supports **remote learning**, minimizing travel and lowering **emissions**.
•Enhances **resource efficiency** through **data-driven planning** and **reporting**.

❖ Impact on Target Audience :

1. Empowers students with verified, institution-backed academic and extracurricular profiles.
2. Boosts employability and higher education prospects through trusted digital portfolios.
3. Improves learning outcomes with AI-driven analytics and personalized mentoring tools.
4. Ensures inclusive access via offline support, local languages, and safe reporting features.
5. Reduces dropout rates by addressing infrastructure gaps, curfews, and gender safety concerns.
6. Simplifies accreditation and scheme access with auto-generated reports and seamless documentation.
7. Strengthens resilience in conflict-affected areas by ensuring continuous, data-driven learning.
8. Fosters community engagement through mentorship, peer networks, and government collaborations.

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RESEARCH AND REFERENCES

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References – Types of Sources

Government Portals – Education schemes, audits, stats
➤ https://jk.gov.in/jammukashmir/education/, https://www.ugc.ac.in/, https://www.education.gov.in ,
Research Papers – Literacy, dropout rates, infrastructure gaps
➤ https://scholar.google.com, https://www.researchgate.net, https://www.jstor.org
Market & Scholarship Data – Fees, aid schemes, enrolment stats
➤ https://data.gov.in, https://scholarships.gov.in , http://naac.gov.in/index.php/en/
Connectivity & Disaster Reports – Internet shutdowns, conflict zones
➤ https://trai.gov.in, https://mausam.imd.gov.in, https://reliefweb.int
AI/ML Tools – Analytics, attendance tracking, portfolio validation
➤ https://www.tensorflow.org, https://scikit-learn.org/stable/, https://www.openai.com
NGOs & Reports – Gender safety, inclusion, field surveys
➤ https://www.unesco.org, https://www.unicef.org, https://www.worldbank.org
Accreditation Guidelines – NAAC SSR, CGPA, audits
➤ http://naac.gov.in/index.php/en/ , https://www.aicte-india.org, https://www.ugc.ac.in
Digital Infrastructure – LMS, ERP, secure platforms
➤ https://moodle.org, https://www.odoo.com, https://firebase.google.com

❖ Comparison with Existing Systems

Features	Our Hub	Others
Verified profiles	✅ Authenticated	❌ Unverified
Multi-language UI	✅ Kashmiri, Urdu, Hindi	❌ Limited
Real-time tracking	✅ Attendance + performance	❌ Attendance only
Schemes access	✅ Integrated (scholarships, loans)	❌ Scattered info
Offline access	✅ Works during internet cuts	❌ Internet-only
NAAC reports	✅ Auto-generated	❌ Manual
Gender safety	✅ Anonymous, safe UI	❌ Neglected
Mentoring & community	✅ Verified networks	❌ Weak forums
AI recommendations	✅ Personalized	❌ None
Security & integration	✅ Encrypted, RBAC, govt link	❌ Weak systems
Disaster support	✅ Attendance recovery	❌ None
Data insights & portfolios	✅ Analytics + shareable profile	❌ Outdated/no tools
Online classes	✅ Supported	✅ Limited
Attendance management	✅ Smart tracking	✅ Basic
UI/UX	✅ Simple, offline-friendly	❌ Internet-dependent

Video link : https://youtu.be/xAEJoBaAy2Y?si=Ua2g2wHExf6W2Z-6